

Longcheng Li

Cambridge, UK | lilongcheng116@gmail.com | longcheng.li

Education

University of Cambridge , Cambridge, UK	Jan 2026 –
• PhD in Computer Science Supervisor: Tom Gur	
University of Chinese Academy of Sciences , Beijing, China	Sep 2022 – Jun 2025
• MS in Computer Science GPA: 3.93/4.0 Supervisor: Jialin Zhang	
University of Chinese Academy of Sciences , Beijing, China	Sep 2018 – Jun 2022
• BS in Computer Science GPA: 3.92/4.0 Rank 2/104	

Conference Publication

*Authors are listed in alphabetical order.

- Cryptomania v.s. Minicrypt in a Quantum World** [🔗]
L. Li, Qian Li, Xingjian Li, Qipeng Liu
Annual International Cryptology Conference (CRYPTO), 2026
- Toward the Impossibility of Perfect Complete Quantum PKE from OWFs** [🔗]
L. Li, Qian Li, Xingjian Li, Qipeng Liu
Innovations in Theoretical Computer Science (ITCS), 2025
- Quantum Byzantine Agreement Against Full-Information Adversary** [📺] [🔗]
L. Li, Xiaoming Sun, Jiadong Zhu
International Symposium on Distributed Computing (DISC), 2024
- How (not) to Build Quantum PKE in Minicrypt** [🔗]
L. Li, Qian Li, Xingjian Li, Qipeng Liu
Annual International Cryptology Conference (CRYPTO), 2024

Journal Publication

- Communication Complexity of Distributed Unitary Synthesis** [🔗]
L. Li, Xiaoming Sun, Jialin Zhang, Jiadong Zhu
IEEE Transactions on Information Theory, 72, 6781 (2026)
- Logical Operations with a Dynamical Qubit in Floquet-Bacon-Shor Code** [🔗]
Xuandong Sun, L. Li (co-first author), Zhiyi Wu, et al.
Physical Review Letters, 135, 220601 (2025)
- Fast exact synthesis of two-qubit unitaries using a near-minimum number of T gates** [🔗]
L. Li, Cheng Guo, Qian Li, Xiaoming Sun
Physical Review A, 107 (4), 042424 (2023)

Teaching

Teaching Assistant of <i>Combinatorics</i>	UCAS, Fall 2023
Teaching Assistant of <i>Introduction to Computer Science</i>	UCAS, Spring 2020

Visiting

Southern University of Science and Technology , Shenzhen, China	Mar 2024 - Jun 2024
• Visiting Student on quantum error correction Advisor: Youpeng Zhong	

Awards

National Scholarship	Nov 2024
Silver Medal, <i>45th International Collegiate Programming Contest Asia East-Continent Final</i>	Apr 2021

Skills

Coding: C/C++, Python

English: IELTS 7.5/9 (Listening 8.5, Reading 8.5, Writing 6.5, Speaking 7.0)